Container Orchestration Using Kubernetes and Apache Mesos

The deployment of containers has changed the way organisations create, ship and maintain applications in real-time. Container orchestration automates the deployment, maintenance, scaling as well as networking of containers. In today's world, where enterprises are required to deploy and manage multiple hosts, container orchestration can be the only rock-solid alternative.

Image Source from www.freepik.com

Container platforms, led by ubiquitous Docker, are now being used to package applications so that they can access a specific set of resources on a physical or virtual host's operating system. In microservice architectures, applications are further broken up into various discrete services that are each packaged in a separate container. The most important advantage of containers to the organisation is adherence to continuous integration and continuous delivery (CI/CD) practices, as containers are scalable. Ephemeral instances of applications or services, hosted in containers, come and go as needed. But, sometimes, scalability becomes an operational challenge. If an enterprise is using ten containers and four to five applications, it's not at all complex to manage the deployment. But if the containers increase to (say) 1000 and applications scale up to 400-500, then management becomes more complex. Container orchestration automates the process in terms of deployment, management, scalability, and even the 24x7 availability of containers.

What exactly is container orchestration?

In order to manage the life cycle of containers, container orchestration is used in varied enterprise environments. It controls and automates the following significant tasks:

- Resource allocation
- Redundancy and 24x7 availability
- Deployment and provisioning
- Health monitoring of all containers and attached hosts
- Real-time addition and removal of containers across the entire infrastructure
- Application configuration

How container orchestration works

The container orchestrator plays an important role in container management and deployment. It helps to manage multiple containers with different versions, network configurations and relationships by scheduling them using its master node.